

稳态网络模拟器使用说明

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稳态网络模拟器是一个用于医学教育的动态反馈系统，用来观察血压、心率、血容量、RAAS、肾脏排泄、呼吸与酸碱、血糖、电解质、组织供氧和乳酸之间的相互耦合。它不是单一指标计算器，而是一个多变量负反馈网络：当你改变一个参数时，系统会先呈现较快的神经反射；随后，体液、内分泌和肾脏调节会逐步参与并增强。为便于教学观察，模拟器将不同生理过程的时间尺度进行了压缩，其中“约 20 个模拟秒后”表示较慢调节开始变得明显，并不对应真实人体内固定的反应时间。若偏离过大、持续存在或干预方向错误，生命稳定度会下降，并可能进入失代偿。本工具仅用于医学教育、病理生理思维训练和课堂演示，不作为临床诊断或治疗依据。



顶部标题区

显示核心说明、制作人信息，并提供语言切换。右侧有挑战场景选择、暂停/继续、归零干预、重新开始等按钮。

干预控制台（左边）

这是主要操作区。每个滑条代表对某个生理变量施加“外源性推动”。

向右拖动表示增加该变量或增强该方向的作用；向左拖动表示降低该变量或施加相反方向。滑条范围是 -100 到 +100。放手后，干预会被系统保持约 20 秒，之后逐步交还给反馈网络并缓慢衰减。

参数分组筛选

控制台上方可按系统筛选变量：全部、心血管、神经体液、肾脏体液、呼吸酸碱、代谢血液。用于在复杂变量较多时快速聚焦某一类调节。

复杂稳态网络

中间网络图显示变量之间的影响关系。节点颜色代表当前偏离方向和程度：蓝/青表示偏低，绿色接近正常，黄/橙/红表示偏高或危险。连线表示变量之间的作用方向，促进和抑制用不同颜色表示；连线越亮，说明该关系在当前状态下影响越明显。点击节点可以在右侧查看该变量的主要输入、主要影响和当前干预值。

生命监测

显示关键指标的实时数值和趋势线，包括 MAP、HR、CO、血容量、GFR、通气、PaO₂、PaCO₂、pH、血糖、血钾、组织供氧等。趋势图适合观察干预后的延迟反应。

生命稳定度

右侧顶部的“生命稳定度”相当于综合风险条。多个指标进入警戒区或危险区、错误干预持续存在、低灌注/缺氧/酸中毒/高钾/低血糖等状态累积时，稳定度会下降。若降至 0，系统判定为稳态失败。

临床病情评估

右侧会根据当前状态识别综合征样模式，例如低血压/休克趋势、低通气/呼吸性酸中毒、低氧血症、钾异常、低血糖、高血糖高渗、高血压容量负荷、乳酸性酸中毒等。每条会提示严重度、有利策略和错误策略。

速度控制

速度 ×1 表示模拟时间接近真实观察节奏（除体液调节外）。提高速度可以加快变化，降低速度便于教学演示和观察反馈过程。

推荐使用流程

先点击“开始模拟”，让系统进入基线状态。**一、扰动：**建议第一次使用时不要同时拖动很多滑条，而是一次只扰动一个变量，例如降低血容量、升高气道阻力、升高血糖或升高代谢需求。**二、观察：**观察网络节点颜色、生命监测趋势和临床病情评估如何变化。**三、干预（备选）：**随后尝试用相反或生理上合理的干预把系统带回稳态。例如，选择“急性失血”后，可以观察血容量下降、静脉回流下降、每搏量和心输出量下降，MAP 下降后交感张力升高、心率和外周阻力代偿升高。若继续降低血容量或压低收缩力，生命稳定度会快速下降；若适度提高血容量、静脉回流、收缩力或外周阻力，系统可能恢复。

一：调节“干预控制台”中的任意参数进行扰动或在疾病挑战模块选择疾病



三：尝试用相反或生理上合理的干预把系统带回稳态

二：观察复杂稳态网络的动态变化（上）、生命监测（下）以及生命稳定度（右）的动态变化

涉及的调控通路

即时环路：血压升高 → 压力感受器增强 → 交感降低、迷走增强 → 心率/外周阻力下降。血压过低则相反。

呼吸环路：CO₂ 升高或 pH 降低 → 化学感受器增强 → 通气升高 → CO₂ 降低，pH 回升。

慢性环路：20 s 后 RAAS、醛固酮、ADH、GFR、尿量与血容量开始显著改变。

极端状态：严重低压、缺氧、酸中毒、高钾、低血糖等不会立刻结束，但会通过乳酸、心肌收缩力、肾小球滤过率和激素环路形成失代偿。

疾病模拟

你还可以选择右上角

选择疾病大类

 并

选择具体疾病

 模拟特定的疾病场景并尝试纠正它们。具体疾病类型如下：

大类	疾病 / 场景	主要病理生理重点
急症与休克	急性失血性休克	失血、低血容量、低灌注与压力反射
	脓毒性分布性休克	血管扩张、微循环供氧障碍、乳酸升高
	过敏性休克伴支气管痉挛	血管扩张、毛细血管渗漏、气道阻塞
	大面积肺栓塞伴阻塞性休克	肺循环阻塞、低氧、有效心排量下降
	张力性气胸伴阻塞性休克	通气受损、静脉回流受阻、低氧
	心包填塞伴阻塞性休克	心室充盈受限、每搏量和心排量下降
心血管	心源性休克	收缩力下降、低心排量、组织低灌注
	急性左心衰伴肺水肿	泵衰竭、肺氧交换受损、低氧
	高血压危象样失衡	外周阻力升高、后负荷和心肾负担
	不稳定性快速心律失常	节律不稳定、充盈减少、有效心排量下降
内分泌与代谢	胰岛素过量致低血糖	胰岛素效应过强、反调节反应、神经低糖风险
	糖尿病酮症酸中毒样失衡	胰岛素不足、高血糖、脱水和代谢性酸中毒
	高渗高血糖状态	重度高血糖、高渗、渗透性利尿
	原发性肾上腺危象	低容量、低血压、低钠、高钾和低血糖
	黏液性水肿昏迷样失衡	低代谢、心动过缓、低通气、低钠
呼吸与酸碱	急性重症哮喘	气道阻塞、低氧、CO ₂ 潴留
	阿片类药物过量致呼吸抑制	中枢呼吸驱动抑制、低通气、高碳酸血症
	COPD 急性加重伴 CO ₂ 潴留	气流受限、慢性/急性高碳酸血症及肾代偿
	低通气性呼吸性酸中毒	肺泡低通气、CO ₂ 潴留、酸血症
	过度通气性呼吸性碱中毒	PaCO ₂ 下降、碱血症
	乳酸性代谢性酸中毒	乳酸生成、碳酸氢盐消耗、代偿性通气
	水杨酸中毒混合性酸碱失衡	呼吸性碱中毒合并代谢性酸中毒
肾脏、水与电解质	高渗性脱水	水分丢失、低容量、高钠和高渗
	SIADH 样低钠血症	ADH 过强、保水、低渗性低钠
	急性肾功能衰竭伴高钾	GFR下降、排钾障碍、容量与电解质失衡
	严重低钾血症	排钾增加、心肌电不稳定和无力
	尿崩症样高钠脱水	ADH 效应不足、多尿、自由水丢失
	急性高盐负荷	钠负荷、渗透压升高和肾脏代偿
血液与氧运输	重度贫血伴组织供氧不足	携氧能力下降；PaO ₂ 正常也可组织缺氧
神经系统急症	癫痫持续状态样代谢危机	高代谢需求、耗氧耗糖、乳酸升高

生命稳定度说明

“生命稳定度”是本模拟器对全身稳态风险的综合评分，范围为 0-100。它不是某一个单独指标，而是把血压、心输出量、血容量、氧合、CO₂、pH、血糖、血钾、组织供氧、乳酸、GFR 等多项变量的偏离程度综合起来计算。当多数指标处于正常或可代偿范围时，生命稳定度会维持在较高水平，甚至逐渐恢复到 100。若某些指标进入警戒区，稳定度可能缓慢下降，表示系统仍有代偿能力，但负担正在增加。若指标进入危险区，尤其是严重低血压、低氧、酸中毒、高钾、低血糖、组织供氧不足或乳酸升高持续存在，生命稳定度会更快下降。需要注意的是，生命稳定度还会受到“干预方向”的影响。正确干预可能减缓下降并帮助系统恢复；错误干预会加速恶化。例如低血容量休克时继续增加尿量或降低血容量，会让稳定度下降更快；低通气伴 CO₂ 潴留时继续降低通气或增加气道阻力，也会加速失代偿。当生命稳定度降至 0 时，表示系统已进入稳态失败/失代偿状态。此时模拟器会暂停并提示主要失衡原因，方便复盘：是哪一组变量最先失衡，干预是否及时，方向是否正确，以及多个反馈环路如何相互放大。

重要解释

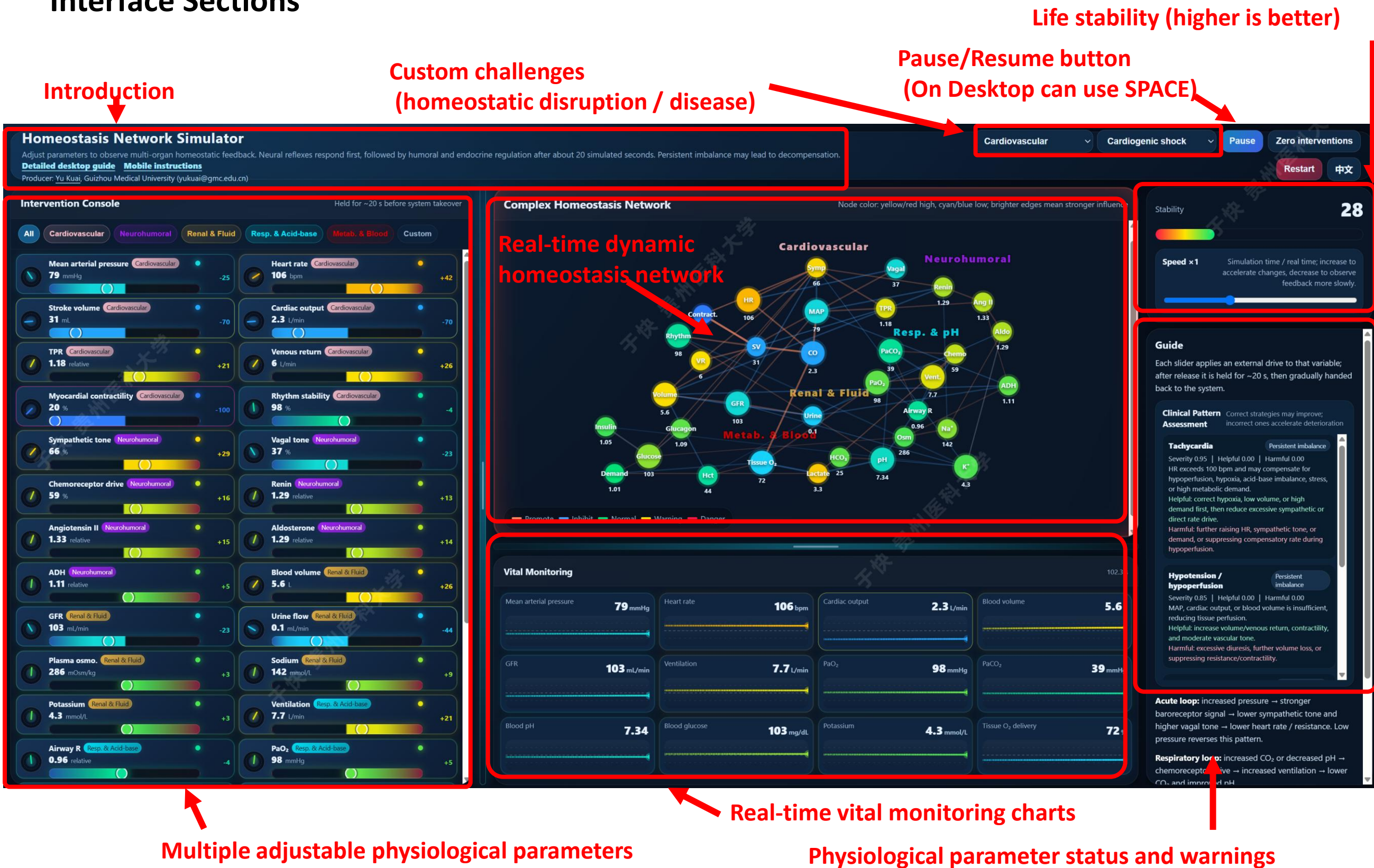
这个模拟器的核心不是“把某个数值调到正常就结束”，而是看系统是否能重新形成稳定反馈。短时间内，一个变量变好可能会让另一个变量变差；例如升高外周阻力可暂时提高血压，但过强阻力可能影响组织灌注。提高通气可改善 CO₂ 和 pH，但若根本问题是低血容量或低心输出量，仅靠通气不能完全恢复组织供氧。

User Guide for the Homeostasis Network Simulator

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The Homeostasis Network Simulator is a dynamic feedback system for medical education. It illustrates the coupling among blood pressure, heart rate, blood volume, the RAAS, renal excretion, respiratory and acid–base regulation, glucose, electrolytes, tissue oxygen delivery, and lactate. It is not a single-variable calculator, but a multivariable negative-feedback network: when a parameter is changed, faster neural reflexes appear first, followed by progressively stronger humoral, endocrine, and renal regulation. To make these processes easier to observe, the simulator compresses physiological time scales for teaching purposes. “After about 20 simulated seconds” means that slower regulatory processes become more apparent in the model; it does not represent a fixed response time in the human body. Large, persistent disturbances or harmful interventions can reduce stability and may lead to decompensation. **This tool is intended solely for medical education, pathophysiology reasoning, and classroom demonstration. It is not intended for clinical diagnosis or treatment decisions.**

Interface Sections



Top Title Area

Displays the core explanation and creator information, and provides language switching. On the right are buttons for selecting challenge scenarios, pause/resume, zeroing interventions, and restarting.

Intervention Console (Left)

This is the main operating area. Each slider represents an “external drive” applied to a physiological variable.

Dragging to the right increases that variable or strengthens that direction of action. Dragging to the left decreases that variable or applies the opposite direction. The slider range is -100 to +100. After release, the intervention is held by the system for about 20 seconds, then gradually handed back to the feedback network and slowly decays.

Parameter Group Filters

The top of the console can filter variables by system: All, Cardiovascular, Neurohumoral, Renal & Fluid, Respiration & Acid–base, and Metabolism & Blood. This helps users focus quickly on one type of regulation when many variables are present.

Complex Homeostasis Network

The central network diagram shows influence relationships among variables. Node colors represent the direction and degree of current deviation: blue/cyan means low, green means near normal, and yellow/orange/red means high or dangerous.

Lines represent the direction of action between variables. Promotion and inhibition are shown with different colors. The brighter the line, the stronger the influence in the current state. Clicking a node lets you view that variable's main inputs, main effects, and current intervention value on the right.

Vital Monitoring

Displays real-time values and trend lines for key indicators, including MAP, HR, CO, blood volume, GFR, ventilation, PaO₂, PaCO₂, pH, blood glucose, blood potassium, tissue oxygen delivery, and others. The trend charts are useful for observing delayed responses after an intervention.

Life Stability

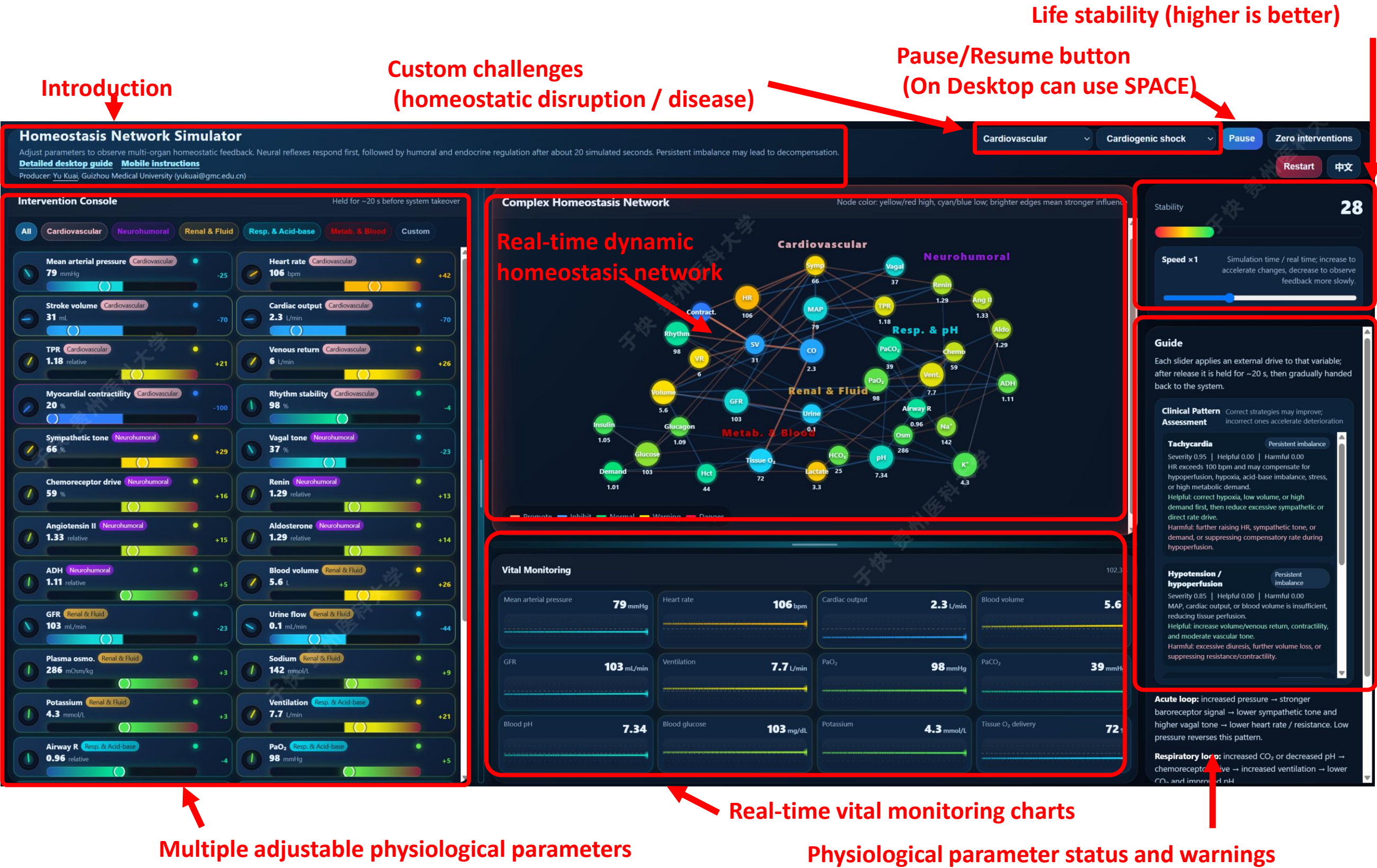
The “Life Stability” indicator at the top right acts as an overall risk bar. Stability decreases when multiple indicators enter warning or danger ranges, when incorrect interventions persist, or when states such as hypoperfusion, hypoxia, acidosis, hyperkalemia, or hypoglycemia accumulate. If it falls to 0, the system is judged to have failed homeostasis.

Clinical Condition Assessment

The right panel identifies syndrome-like patterns according to the current state, such as hypotension/shock tendency, hypoventilation/respiratory acidosis, hypoxemia, potassium abnormalities, hypoglycemia, hyperglycemic hyperosmolar state, hypertension/volume overload, and lactic acidosis. Each item shows severity, helpful strategies, and harmful strategies.

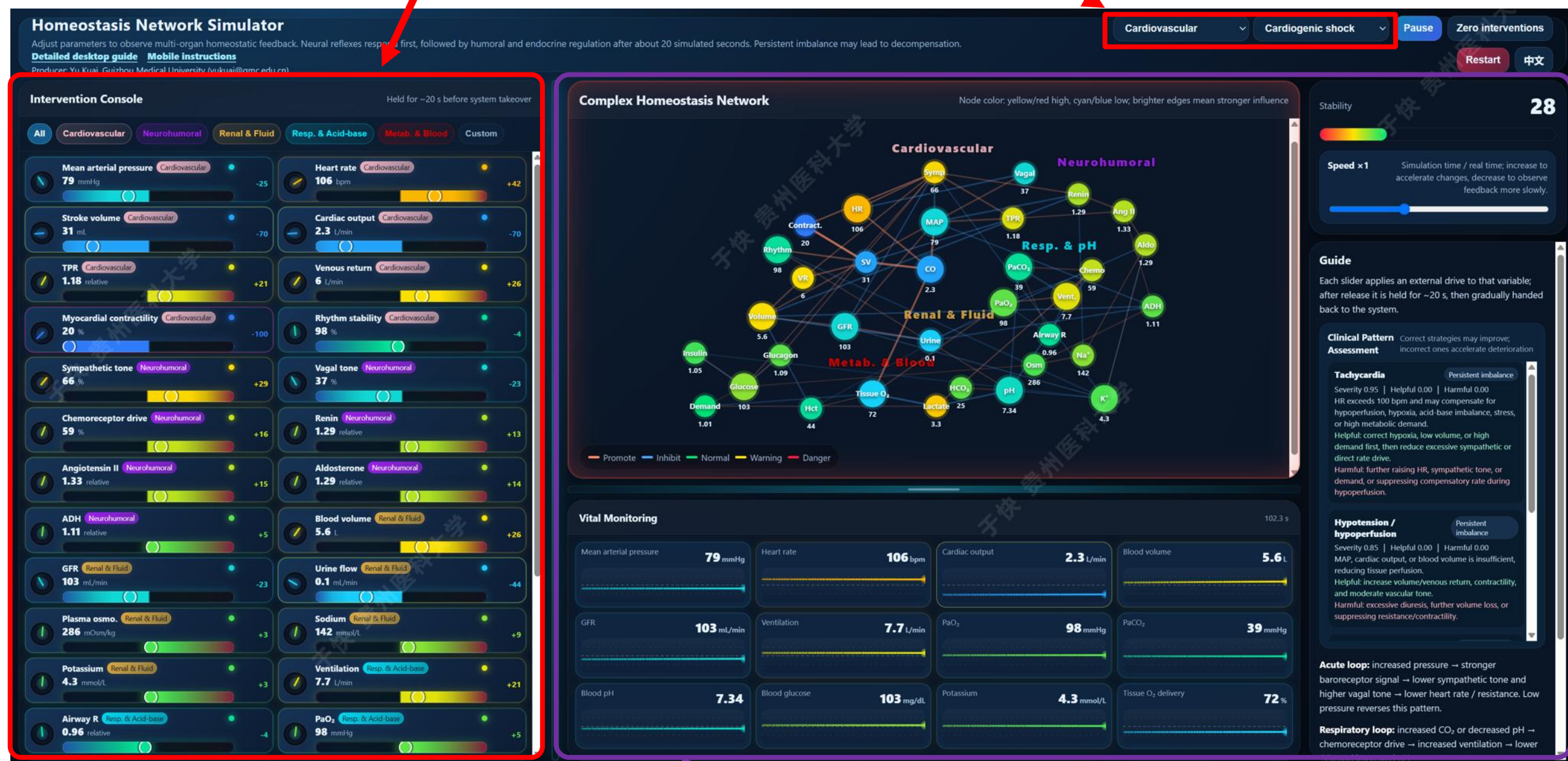
Speed Control

Speed ×1 means the simulation time is close to the real observation pace, except for fluid regulation. Increasing the speed accelerates changes; decreasing it makes feedback easier to demonstrate and observe.



Recommended Workflow

1. Disturb any parameter in the Intervention Console or select a challenge scenario



3. Optionally bring homeostasis back by Intervening the parameters

2. Observe changes in the network (upper panel), vital parameters (bottom) and stability (right)

First click “Start Simulation” to bring the system into its baseline state.

1. Disturbance:

For first-time use, avoid dragging many sliders at once. Instead, disturb one variable at a time, such as decreasing blood volume, increasing airway resistance, increasing blood glucose, or increasing metabolic demand.

2. Observation:

Observe how the network node colors, vital monitoring trends, and clinical condition assessment change.

3. Intervention (optional):

Then try to bring the system back to homeostasis using an opposite or physiologically reasonable intervention.

For example, after selecting “Acute Hemorrhage,” you can observe decreased blood volume, decreased venous return, decreased stroke volume and cardiac output, and then increased sympathetic tone, heart rate, and peripheral resistance as MAP falls. If blood volume continues to decrease or contractility is suppressed, life stability will fall rapidly. If blood volume, venous return, contractility, or peripheral resistance is increased appropriately, the system may recover.

Feedback Loop/Compensation involved

Acute loop: increased pressure → stronger baroreceptor signal → lower sympathetic tone and higher vagal tone → lower heart rate / resistance. Low pressure reverses this pattern.

Respiratory loop: increased CO₂ or decreased pH → chemoreceptor drive → increased ventilation → lower CO₂ and improved pH.

Chronic loop: after 20 s, RAAS, aldosterone, ADH, GFR, urine flow, and blood volume become prominent.

Extreme states: severe hypotension, hypoxia, acidosis, hyperkalemia, or hypoglycemia may trigger decompensation through lactate, contractility, GFR, and hormonal loops.

Disease/Disorder Scenarios

You can also

Select major disorder

 and

Select specific disorder

 in the upper-right corner to observe and try to correct them.

The scenarios are as follows:

Major category	Disorder / scenario	Main pathophysiologic focus
Emergency & Shock	Acute hemorrhagic shock	Blood loss, hypovolemia, hypoperfusion, baroreflex
	Septic distributive shock	Vasodilation, microcirculatory oxygen failure, lactate
	Anaphylactic shock with bronchospasm	Vasodilation, capillary leak, airway obstruction
	Massive pulmonary embolism with obstructive shock	Pulmonary vascular obstruction, hypoxemia, reduced effective output
	Tension pneumothorax with obstructive shock	Impaired ventilation, obstructed venous return, hypoxemia
	Cardiac tamponade with obstructive shock	Restricted filling, reduced stroke volume and cardiac output
Cardiovascular	Cardiogenic shock	Contractile failure, low output, tissue hypoperfusion
	Acute left-heart failure with pulmonary edema	Pump failure, impaired gas exchange, hypoxemia
	Hypertensive-crisis pattern	Raised resistance, afterload, cardio-renal strain
	Unstable tachyarrhythmia	Rhythm instability, reduced filling, low effective output
Endocrine & Metabolic	Insulin-induced hypoglycemia	Excess insulin effect, counter-regulation, neuroglycopenia
	Diabetic ketoacidosis pattern	Insulin deficiency, hyperglycemia, dehydration, acidosis
	Hyperosmolar hyperglycemic state	Severe hyperglycemia, hyperosmolality, osmotic diuresis
	Primary adrenal crisis	Volume depletion, hypotension, hyponatremia, hyperkalemia, hypoglycemia
	Myxedema-coma pattern	Low metabolism, bradycardia, hypoventilation, hyponatremia
Respiratory & Acid–Base	Acute severe asthma	Airway obstruction, hypoxemia, CO ₂ retention
	Opioid overdose with respiratory depression	Suppressed respiratory drive, hypoventilation, hypercapnia
	COPD exacerbation with CO ₂ retention	Airflow limitation, acute-on-chronic hypercapnia, renal compensation
	Hypoventilation / respiratory acidosis	Alveolar hypoventilation, CO ₂ retention, acidemia
	Hyperventilation / respiratory alkalosis	Low PaCO ₂ , alkalemia
	Lactic metabolic acidosis	Lactate generation, bicarbonate consumption, compensatory ventilation
	Salicylate toxicity with mixed acid–base disturbance	Respiratory alkalosis plus metabolic acidosis
Renal, Fluid & Electrolytes	Hyperosmotic dehydration	Water loss, hypovolemia, hypernatremia, hyperosmolality
	SIADH-like hyponatremia	Excess ADH, water retention, hypotonic hyponatremia
	Acute renal failure with hyperkalemia	Reduced GFR, impaired potassium excretion, volume/electrolyte imbalance
	Severe hypokalemia	Potassium loss, electrical instability, weakness
	Diabetes-insipidus pattern with hypernatremic dehydration	Insufficient ADH effect, polyuria, free-water loss
	Acute high-salt load	Sodium load, raised osmolality, renal compensation
Hematology & Oxygen Transport	Severe anemia with impaired oxygen delivery	Reduced oxygen carriage despite potentially normal PaO ₂
Neurologic Emergencies	Status-epilepticus metabolic crisis	High metabolic demand, oxygen/glucose consumption, lactate rise

Explanation of Life Stability

“Life Stability” is the simulator's comprehensive score for whole-body homeostatic risk, ranging from 0 to 100. It is not a single indicator. Instead, it integrates the degree of deviation across multiple variables, including blood pressure, cardiac output, blood volume, oxygenation, CO₂, pH, blood glucose, blood potassium, tissue oxygen delivery, lactate, and GFR.

When most indicators remain within normal or compensable ranges, life stability stays high and may gradually recover to 100. When some indicators enter the warning range, stability may decline slowly, indicating that compensation is still possible but the system burden is increasing. When indicators enter the danger range, especially if severe hypotension, hypoxia, acidosis, hyperkalemia, hypoglycemia, inadequate tissue oxygen delivery, or elevated lactate persists, life stability falls more rapidly.

It is important to note that life stability is also affected by the direction of intervention. Correct interventions may slow the decline and help the system recover. Incorrect interventions accelerate deterioration. For example, during hypovolemic shock, continuing to increase urine output or further decreasing blood volume will make stability fall faster. During hypoventilation with CO₂ retention, further reducing ventilation or increasing airway resistance will also accelerate decompensation.

When life stability falls to 0, the system has entered homeostatic failure/decompensation. At this point, the simulator pauses and shows the main causes of imbalance, making it easier to review which group of variables first became unstable, whether the intervention was timely, whether its direction was correct, and how multiple feedback loops amplified one another.

Important Explanation

The core purpose of this simulator is not simply to “adjust a number back to normal,” but to determine whether the system can re-form a stable feedback state. In the short term, improving one variable may worsen another. For example, increasing peripheral resistance may temporarily raise blood pressure, but excessive resistance may impair tissue perfusion. Increasing ventilation can improve CO₂ and pH, but if the root problem is low blood volume or low cardiac output, ventilation alone cannot fully restore tissue oxygen delivery.